Senior High School

EARTH SCIENCE
Quarter 1- Module

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Earth Science – Grade 11
Alternative Delivery Mode
Quarter 1 – Module
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Table of Contents

WEEK 1: The Structure of the Earth	
Day 1: Characteristics of the Earth that Supports Life	1
Day 2: Earth's Subsystems	3
Day 3: Physical & Chemical Properties of Rock-Forming Minerals	5
WEEK 2: Earth Materials and Processes	
Day 1-2: Three Types of Rocks	9
Day 3: Useful Minerals	13
WEEK 3: Fossil Fuels Formation and Ore Minerals	
Day 1-2: How do we extract minerals?	18
Day 3-4: Bread Fossils: Discover the Origin of Fossil Fuels	22
WEEK 4: Energy and Water Resource	
Day 1: Energy from Heat Underneath	27
Day 2: Water Resources	31
WEEK 5: Activities Affecting Water and Soil Resources	
Day 1: Activities Affecting Water Quality and Availability	34
Day 2: Activities Affecting Quality and Quantity of Soil	36
WEEK 6: Soil Conservation and Protection, Waste Generation and Classification	
Day 1: Methods of Controlling Soil Erosion	40
Day 2: Waste Classification and Generation	43
WEEK 7: Waste Generation and Management	
Day 1-2: Waste Generation & Management	47

The Structure of the Earth
Quarter 1 – Week 1

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the topics about the Earth's structure. The scope of this module permits it to use in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged that follows the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into three parts:

- Module 1 – Characteristics of Earth that Support Life
- Module 2 – Earth's Subsystems
- Module 3 – Physical & Chemical Properties of Rock-forming Minerals

After going through this module, you are expected to:

1. Explain the life-supporting characteristics of Earth.
2. Describe the four subsystems of the Earth.
3. Identify the physical and chemical properties of rock-forming minerals.

What I know

LET'S TRY! (PRE-ASSESSMENT)

Directions: Fill in the K-W-H-L Chart below to assess your prior knowledge and understanding of the topic, life-supporting characteristics of planet earth, Earth's subsystems, and properties of rock-forming minerals.

K	W	H	L
What do I know?	**What do I want to find out?**	**How can I found out what I learn?**	**What did I learn?**

Day 1 — Characteristics of the Earth that Supports Life

Earth is the planet we live in, and it is the third planet that revolves around the sun. Our habitat comprises life-supporting components, such as the presence of free oxygen in the atmosphere, water in our oceans, and other elements that make Earth habitable.

At the end of this lesson, you are expected to:

1. describe the characteristics of Earth that are necessary to support life. **[S11/12ES-Ia-b-3]**

What's In

Earth is a habitable planet in our solar system and galaxy. Our planet is a product of cosmic evolution 4.6 billion years ago. Whenever scientists discover an Earth-like planet, they usually compare it to the attributes of the Earth which support life. This module will help you understand the life-supporting characteristics of our home planet.

What is It

Life Supporting Characteristics of Planet Earth

1. The moon: The Earth has a slight tilt which causes drastic shifts in climate throughout thousands of years. **2. Stable rotation:** The regularity and frequency of day and night on planet Earth go far to prevent extreme temperatures and encourage life. **3. A magnetic field:** Our planet is blessed with a strong and stable magnetic field that staves off the cosmic rays and solar flares that would otherwise fry the planet now and then. **4. Dynamic geology:** The cloud of gas and dust that eventually coalesced into the Earth contained enough radioactive elements to keep the core of the planet churning merrily for billions of years. **5. Atmosphere:** The occurrence of the correct percentage of life-sustaining gases such as nitrogen, hydrogen, and others enable the living organisms to survive. One of the essentials gasses present in the Earth's atmosphere is oxygen which builds the ozone layer. **6. Isolation:** Venus and Mars are close to the Earth, but our solar system as a whole is in the middle of nowhere. And because we are far from the major spiral arms of the Milky Way, we're in a lot less danger of running afoul of some greater star's gravitational pull (among other hazards). **7. Long-lived sun:** Our sun is a yellow dwarf, a relatively rare type of star that's both small and stable. It also has a long life, and probably won't start to fizzle out for another five billion years or so. **8. Plenty of time:** The sun has about five billion years left, and the Earth itself is around four and a half billion years old. But life only arrived in that last half-billion. We're here because our planet was sturdy enough to last until the first signs of life appeared. **9. Gas-giant neighbors:** The intense gravity of the neighboring planets such as Jupiter conveniently catch wayward asteroids and comets, making events like the one that killed the dinosaurs a rarity.

What is It

Materials: Worksheets & Pen

Procedure:

1. **Scenario:** The aliens from planet X from the nearby galaxy plans to invade our planet for them to continue their race. The aliens found out that the Earth is a habitable planet that can sustain their life. Using the given scenario, you are assigned to create a comic strip relating your understanding of the life-sustaining characteristics of planet Earth. You may place your comic strip in clean bond paper.
2. To get good ideas on the storyline and dialogue in your comic strip, you may ask someone from your family to share their concept of an alien invasion. You may collaborate with them for better output. Place your answer in a long bond paper.
3. Using social media, you may share your ideas with your groupmates by exchanging thoughts and concepts about the topic and your draft. Be careful not to copy someone else work. Let us make it your original work.

Guided Questions:

1. What are the characteristics of Earth that support life?

2. How does your learning about the topic can help others become aware of the importance of planet Earth to us?

What I have Learned

In your own words, explain how planet Earth supports life.

What I can do

If one of the Earth's life-supporting characteristics deteriorates, what could happen to the human race?

Assessment

Directions: Write True if the statement is correct and False if not. Write your answer in the space provided.

_____1. The moon gives a stabilizing effect on our orbit which, makes our climate steady. As the moon rotates around the Earth it causes tides where most biologists think that life begins with tidal pools.
_____2. The presence of gas giants such as Jupiter helps the Earth, to have stable gravitational control in our solar system.
_____3. Our sun is a yellow dwarf star. It is considered the main source of energy on Earth.
_____4. Earth's atmosphere is comprised of free oxygen which builds up the ozone layer.
_____5. The Earth is reliant on the magnetic field of the inner planets because it has a weak gravity.

Day 2 — Earth's Subsystems

Earth comprises 71% of water and 29% of the landmass. 3.5% of Earth's water is freshwater and, 96.5% of it contains ocean saltwater. According to NASA, the gasses in the Earth's atmosphere include 78% percent nitrogen, 21% oxygen, 0.93% argon, 0.04% carbon dioxide, and a trace amount of neon, methane, helium, krypton, and hydrogen, as well as water vapor. This composition has fair interaction with one another that provides the basic needs of the living creatures on our planet.

At the end of this lesson, you are expected to:

1. explain that the Earth consists of four subsystems, across whose boundaries matter and energy flow. **[S11/12ES-Ib-4]**

What's In

We discussed from the previous modules the different life-supporting characteristics of planet Earth. However, trying to understand how this planet we call home works can seem confusing because many things are going on at once! Scientists break down the way the Earth system works into several subsystems: the atmosphere, biosphere, geosphere (or lithosphere), and hydrosphere. It is

important to remember that the subsystems do not work in isolation – each one depends on and interacts with the other three. The next lesson will be teaching you the Earth's subsystem and how their interaction affects us.

What is It

The Earth's Subsystem

1. Atmosphere - The atmosphere is the layer of *gases* that surround the Earth. Compares to the size of the Earth, it is a thin layer, composed primarily of *nitrogen* and *oxygen* with small amounts of *carbon dioxide* and other gases. The atmosphere is significant for several reasons: it protects the Earth from incoming solar rays; it circulates the gases that plants and animals need to survive; it is also responsible for our *weather.* **2. Biosphere** - The biosphere consists of all the living organisms on Earth. Their *habitats* extend from the upper areas of the atmosphere, to deep in the ground, to the bottom of the ocean – any place that life can exist. **3. Geosphere** - The geosphere is the physical Earth – the rock, *magma,* and soil. The geosphere extends from the center of the Earth (the core, *mantle,* and *crust*) to the dust in the atmosphere and even includes the *sediments* found in the oceans. **4. Hydrosphere** - The hydrosphere is all the water held on the Earth – water molecules in the air, icebergs and glaciers, *groundwater,* lakes, rivers, and oceans.

What's More

Materials: Worksheets & Pen

Procedure:

1. You are tasked to create a concept map about the interconnection of the Earth's subsystems. It is expected that you make your output creative by using various designs on your work. Place your work in a clean bond paper.
2. Conceptualizing your map is challenging! You may seek help from your teacher and classmates to improve the content of your concept map.
3. To create awareness, you may share your concept map online. You may create a creative video presentation to make it easier to understand by incorporating a voice that explains the idea of your output.

Guided Questions:

1. How do the Earth's subsystems function?
2. If one of the subsystems has been damaged, what will happen to Earth?

What I have Learned

Directions: Complete the table below by describing the function of each subsystem of the Earth.

Subsystem	Function
Atmosphere	
Biosphere	
Geosphere	
Hydrosphere	

What I can do

Directions: Briefly answer the given question below.

1. What are the adverse effects of pollution on the Earth's subsystems?

Assessment

Directions: Identify what is being described in the statement. Choose from the word bank and write your answer on the space provided.

Word Bank				
-Hydrosphere	-Mesosphere	-Crust -Core	-Troposhere	-Mohorovicic
Discontinuity	-Stratosphere	-Magnetosphere	-Mantle	-Lithosphere
-Mesosphere				

_____1. The complex middle region beneath the crust.
_____2. The innermost layer of the Earth that comprises of iron and nickel.
_____3. The outermost layer of the geosphere.
_____4. The sharp boundary between the crust and mantle.
_____5. It is the lowest layer of the atmosphere that contains 75% of all the air and water vapor.
_____6. The atmospheric layer that contains much of the ozone that protects the earth from harmful radiation.
_____7. The region above the stratosphere where the temperature decreases with height, reaching a minimum of about -90°C.
_____8. The outer region of the atmosphere that surrounds the earth, where charged particles spiral along the magnetic field lines.
_____9. The subsystem that consists of all the living organisms on Earth.
_____10. The subsystem of the Earth that is primarily made up of water.

Day 3 — Physical & Chemical Properties of Rock-Forming Minerals

The solid earth is made of rocks which are made of minerals. To understand rocks, you need to become familiar with minerals and how they are identified. Minerals are naturally occurring, inorganic solids with definite chemical composition and a crystal lattice structure. Although thousands of minerals on earth have been identified based on their physical and chemical characteristics, having a basic understanding of it can help us realize how important it is to protect our natural resources to help our planet serve its purpose to all living kinds.

At the end of this lesson, you are expected to:

1. identify the common rock-forming minerals using their physical and chemical properties. **[S11/12ES-Ib-5]**

What's In

After understanding the different subsystems of planet earth, we proceed to discuss some of the properties we can see from rocks. As part of the lithosphere, rocks are made based on various processes that make them unique based on the process and characteristics that make it distinct from another. By the end of this module, you will learn the physical and chemical properties of rock-forming minerals.

What is It

Mineralogists are scientists who study minerals. One of the things mineralogists must do is identify and categorize minerals by its physical and chemical characteristics.

Physical Properties of Rock-Forming Minerals

1. **Color** - Some minerals come in many different colors. Quartz, for example, may be clear, white, gray, brown, yellow, pink, red, or orange.
2. **Luster** describes the reflection of light off a mineral's surface.
3. **Streak** is the color of a mineral's powder.
4. **Specific Gravity** of a substance compares its density to that of water. Denser substances have higher specific gravity.
5. **Hardness** is the strength in which a mineral resists its surface being scraped or punctured.
6. **Cleavage** is the tendency of a mineral to break along certain planes to make smooth surfaces.
7. **Fracture** is a break in a mineral that is not along a cleavage plane.
8. **Crystal Shape** - All minerals are crystalline, but only some have the opportunity to exhibit the shapes of their crystals.

Chemical Properties of Rock-Forming Minerals

Minerals are classified according to their chemical properties. Except for the native element class, the chemical basis for classifying minerals is the anion, the negatively charged ion that usually shows up at the end of the chemical formula of the mineral. For example, the sulfides are based on the sulfur ion, S^{2-}. Pyrite, for example, FeS_2, is a sulfide mineral. In some cases, the anion is of a mineral class is polyatomic, such as $(CO_3)^{2-}$, the carbonate ion. The major classes of minerals are silicates, sulfides, carbonates, oxides, halides, sulfates. Phosphates and native elements.

For additional lecture, access these links:

https://fac.ksu.edu.sa/sites/default/files/geo_221-unit-2_0.pdf
https://courses.lumenlearning.com/wmopen-geology/chapter/outcome-identifying-minerals/

What's More

Materials: Worksheet, pen, Iron nail, Silver ring, Rock (Any type), Talcum powder, Copper wire, Pencil lead, Gold ring

Procedure:

- Observe the given materials based on the physical properties of minerals. Answer the table below.

Material	Properties of Minerals Observed
Ex. Diamond	It was hard to pound, sparkling and white.
Iron Nail	
Silver Ring	
Gold Ring	
Rock	
Talcum Powder	
Copper Wire	
Pencil Lead	

Guided Questions:

1. What are the common properties of minerals you observed in the given materials?
2. Do the properties of minerals give more value to a specific material? Why?

Directions: Identify the chemical formula of the given chemical property of rock-forming minerals.

Property	Chemical formula
1. Silicates	
2. Sulfides	
3. Carbonates	
4. Oxides	
5. Halides	
6. Sulfates	
7. Phosphate	
8. Native Element	

What I have Learned

Directions: Complete the table by briefly describing the physical and chemical properties of rock-forming minerals. You may perform the activity by sharing your ideas with your classmate online.

Physical Property	Description	Chemical Property	Major Chemical Component
Color		Silicates	
Luster		Sulfides	
Steak		Carbonates	
Specific Gravity		Oxides	
Hardness		Halides	
Cleavage		Sulfates	
Fracture		Phosphate	
Crystal Shape		Native Element	

What I can do

Procedure:

1. You are assigned to create a slogan about the importance of minerals on our planet. You may seek help from your friends in creating your slogan for correct grammar and rhyme.
2. You may share your slogan with your neighbors by posting it outside your home.

Guided Questions:

1. What is the importance of understanding the physical and chemical properties of rock-forming minerals?

Assessment

Directions: Identify what is being described. Choose your answer on the word bank and write your answer on the space provided.

Word Bank				
-Cleavage Luster	-Radioactivity	- Oxides	-Phosphate	-Sulfides
-Fracture	-Streak	-Hardness	- Fluorescence	

_____1. The unusual property of mineral that gives off radiation that can be measured with Geiger counter.
_____2. The reflection of light off a mineral's surface.
_____3. The property of mineral with chemical of base of oxygen anion, O^{2-}.
_____4. The property of mineral with chemical base of polyatomic phosphate ion, $(PO_4)^{3-}$, as the anion.
_____5. The color of a mineral's powder.
_____6. The unusual property of mineral that glows under ultraviolet light.
_____7. The property of mineral with chemical base of sulfide ion, S^{2-}.
_____8. The tendency of a mineral to break along certain planes to make smooth surfaces.
_____9. The break in a mineral that is not along a cleavage plane.
_____10. The strength of the mineral that resist its surface to be scraped or punctured.

Additional Activities

Based on what you have learned from this week's lesson, you are task to create a graphic organizer about the Earth's Structure. Make your work colorful and place it on clean bond paper.

REFERENCES

Creative Commons M.Alan Kazlev, "Structure of the Earth," Modified March 21, 2020. http://palaeos.com/earth/geosphere/structure.html

Lumen, "Identifying Minerals." Accessed on May 28, 2020. https://courses.lumenlearning.com/wmopen-geology/chapter/outcome-identifying-minerals/

NIWA, "Layers of the Atmosphere.' Accessed on May 27, 2020. https://niwa.co.nz/education-and-training/schools/students/layers

Science Learning Hub, "Earth System." Accessed on May 28, 2020. https://www.sciencelearn.org.nz/resources/720-earth-system

Reuben, Westmaas, "9 Things That Make Earth the Perfect Place for Life." August 01, 2019. https://www.discovery.com/science/Earth-Perfect-Place-for-Life

Earth Materials and Processes
Quarter 1 – Week 2

What I Need to Know

This module was designed and written for you. It was developed to help you learn more about Earth materials and processes. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course.

The module is divided into two lessons, namely:

Lesson 1 – Three Types of Rocks
Lesson 2 – Useful Minerals

After going through this module, you are expected to:

1. Describe the characteristics of the three types of rocks.
2. Classify rocks into igneous, sedimentary, and metamorphic.
3. Identify the minerals that are important to society.

What I know

LET'S TRY! (PRE-ASSESSMENT)
Directions: Fill in the K-W-L Chart below with what you **K**now about the topic, what you **W**ant to know, and what you've **L**earned to assess your prior knowledge and understanding of the lesson, Earth Materials, and Processes.

K	W	L
What do I know?	**What do I want to know?**	**What did I learned?**

Day 1 & 2 — Three Types of Rocks

Earth is a dynamic planet and is constantly changing. These changes are brought by processes happening inside and outside the surface of our planet such as erosion, weathering, landslide, volcanic activity, earthquakes, and movements of plates. Such changes had persistently shaped and reshaped our planet since its beginning approximately 4.5 billion years ago.

At the end of this unit, you are expected to:

1. classify rocks into igneous, sedimentary, and metamorphic**.**
   (S11ES-Ic-6)

What's In

You have learned from the previous week that the earth is a very intricate system. But, did you know? There are only four known components of earth that are considered as the building blocks of life. These components are referred to as Earth's materials and include minerals, rocks, soil, and water.

In this lesson, you will learn more about the three types of rocks, namely: igneous, sedimentary, and metamorphic. The place and condition for their formation will also be thoroughly discussed.

What's New

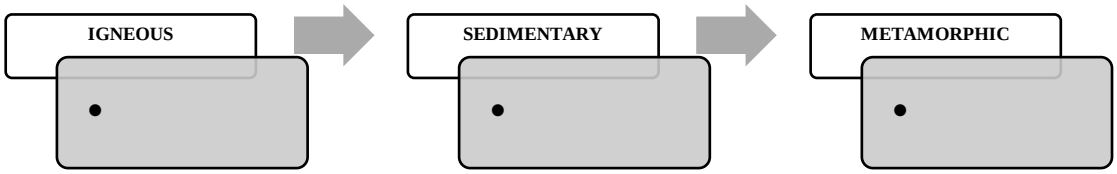
Rocks are naturally occurring resources that are typically a combination of common minerals. It has three main kinds or classes in which the location and process of formation were used as the basis. These three main clusters are known as igneous, sedimentary, and metamorphic rocks. However, despite being classified into diverse families, rocks can be changed from one type to another. The measures and settings necessary for these transformations and changes in rocks are best described by the Rock Cycle Diagram.

IGNEOUS ROCK • Its name was derived from the Latin word “ignis” which means fire. This type of rock is formed when lava or magma cools and solidifies. • The following are the basic characteristics of igneous rock: (1) it normally contains no fossils; (2) it rarely reacts with acids; (3) it usually has no layering; (4) hard in nature, and; (5) consists of crystals.
SEDIMENTARY ROCK • This type of rock is formed when sediments of pre-existing rock or pieces of once-living organisms are deposited near bodies of water as a product of weathering and erosion. • The following are the basic characteristics of sedimentary rock: (1) these rocks are soft; (2) they form layers; (3) they do not contain crystals; (4) they contain fossils and pieces of rocks cemented together, and; (5) they react with strong acids.
METAMORPHIC ROCK • The name of this rock means "change in form". This type of rock started as igneous or sedimentary, but due to natural agencies such as heat and pressure, it substantially changed into a new type of rock. • The following are the basic characteristics of metamorphic rock: (1) they have alternate bands of dark and light minerals; (2) it has ribbin-like layers, and; (3) considered as the hardest rock.

What is It

PART A: Directions: Classify the given rocks within the three main classes using the provided key characteristics by writing their rock code under the correct columns or headings. *(6 points)*

ROCK CODE	KEY CHARACTERISTICS
ROCK A	This rock has **ribbon-like layers**. It forms from other rocks that have been squeezed and heated for a long time deep within the earth’s crust.
ROCK B	This rock has **gas bubbles**. These bubbles form when hot lava from a volcano contains gases that escape from the rock as the lava cools.
ROCK C	Sand and pebbles are visible in this rock. It is made up of pebbles, stones, and smaller particles pressed together by the action of waves or water. The rock is often found in large expanses or beds.
ROCK D	This rock has **crystals**. It forms when **limestone** is pushed down into the earth and subjected to intense heat and pressure for a long period.
ROCK E	This rock has a **glassy surface**. It originated from lava that cools so quickly that no crystals can form on its surface. It can be found near volcanic lava flows.
ROCK F	You can see **fossils** in this rock. These fossils were formed when seashells and skeletons of marine animals were compressed into the seafloor, along with other sediments. This rock is often found near oceans and lakes.



PART B: *(15 points)*

1. Observe the pictures of three rock samples.
2. Identify which rock is igneous, sedimentary, and metamorphic.
3. Apply what you have learned from this module by completing the table below.

_____ _____ _____

Figure 1: Rock Samples
Photo Credits: https://www.rocksforkids.com/

Guide Questions:

Type of Rock	How does this rock form?	What are the characteristics of this type of rock?	Where can you find this type of rock?	Examples
Igneous				
Sedimentary				
Metamorphic				

What's More

Directions: Complete the paragraph below. *(10 points)*

Rocks on earth can be classified into _____, _____ and _____. Igneous rocks are formed when _____ rock cools and _____. Sedimentary rocks originate when _____ of pre-existing rocks settle out of water or air as products of _____ and erosion. They accumulate in _____. Metamorphic rocks on the other hand, result when existing rocks are changed by _____ and _____.

What I have Learned

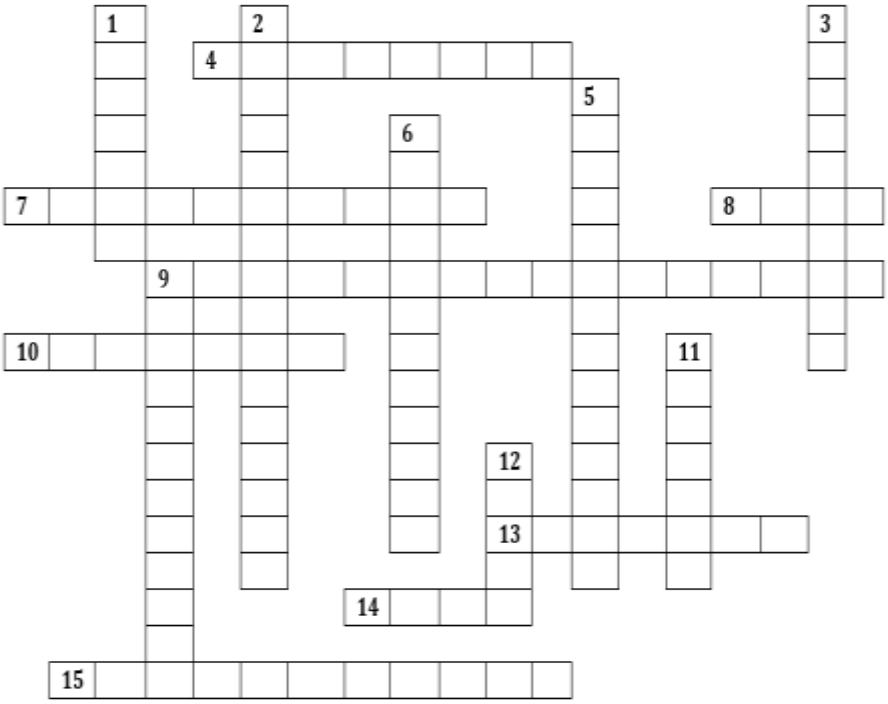
I have learned that… _____

What I can do

Why do geodetic and civil engineers need to know about rocks? *(5 points)*

Assessment

Directions: Complete the crossword puzzle by identifying the term described in each statement. Answers with more than one word will have a space between each word in the puzzle boxes. *(16 points)*



ACROSS		DOWN	
4	Fragments of rock or naturally occurring materials	1	An inorganic solid with a crystalline structure and definite chemical composition
7	Breaking of rocks brought by water, wind, and ice	2	The types of rocks formed from compacted surface sediments
8	A solid material made of minerals	3	The process of formation breakdown and reformation of rocks
9	Rocks changed by extreme pressure and heat	5	The process of layering sedimentary rocks
10	The feeling of a surface of a rock	6	The type of rock formed from the cooling of magma or lava
13	The branch of science that deals with the study of the structure of the earth	9	Changes in the structure of something
14	Magma that reaches the earth's surface	11	The process wherein wind and water move sediment to new places
15	The chemical make-up of a rock	12	The molten rock beneath the earth's surface

Day 3	Useful Minerals

98.5% of the Earth's crust is made up of only eight elements. These are oxygen, silicon, aluminum, iron, calcium, potassium, sodium, and magnesium. These elements are the same materials that make up most minerals.

Minerals are naturally occurring inorganic solids with an orderly crystalline structure and a definite chemical composition.

At the end of this unit, you are expected to:

1. identify the minerals important to the society **(S11ES-Ic-7)**

What's In

You have learned from the previous lesson that rocks constantly change due to the occurrence of natural processes that are taking place all the time. They are made up of various mineral grains combined in different ways. You also learned from the first module that rock-forming minerals are grouped according to their different physical and chemical properties.

In this lesson, the importance and uses of different minerals will be presented comprehensively to make you realize that minerals are not just found in soil but in almost every product you use at home.

What's New

From the roughly 3,800 named minerals in the world, forty of them are used in our daily lives. Some of the names are common and well-known while others are quite unfamiliar but helps in creating something we cannot live without. Some examples of minerals and their importance to society are the following:

ALUMINUM	•Used in making automobiles and airplanes. It is also used in canning industries.
ANTIMONY	•Used as flame retardants, ammunition and automotive batteries.
BISMUTH	•Used in pharmaceutical products and chemical industry as ceramics and paint.
COPPER	•Used in electric cables and wires, switches, electronic components, coins and jewelries.
FELDSPAR	•Used in glass and ceramics industry
GOLD	•Used in destistry and medicine, jewelry and arts, medallions and coins.
GRAPHITE	•Used as dry lubricant, steel hardener, brake linings and production of lead pencils.
HALITE	•Used in table salt, food seasoning and food preservatives.
QUARTZ	•Used to manufacture glass, paints, abrasives and precision instruments.
TALC	•Used in the production of paper, baby powder, deodorant and make-up.

What is It

Directions: Unscramble each word in Column B to find the names of common minerals that come from the earth described in Column A. Write your answer in Column C and match each mineral with the object on which you can find them in Column D. *(12 points)*

Column A	Column B	Column C	Column D
You can find this in jewelry.	imadond		
You can find this in wedding rings.	olgd		
You can find this in a watch.	tzqrua		
You can find this in some forks, knives and spoons.	vseilr		
You can find this at the end of a pencil.	pihtegar		
You can find this in a penny.	poerpc		

What's More

Directions: Your bathroom is made of many mineral resources that it is practically a mine in disguise. Unscramble the words below to identify some of the many mineral products in a bathroom. In the column on the right, write the name of the main mineral ingredient(s) of that product like iron, copper, quartz sand, zinc, clay minerals, etc. The first one is done for you. *(18 points)*

ehwosrhdae	s h o w e r h e a d	iron	tapthotose	t __ __ __ __ __ __ __ __ __	_____
rorirm	m __ __ __ __ __	_____	naf	f __ __	_____
rheydrair	h __ __ __ __ __ __ __ __	_____	lotiet	t __ __ __ __ __	_____
fetcau	f __ __ __ __ __	_____	skin	s __ __ __	_____
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What I have Learned

Directions: Answer the following questions briefly. *(10 points)*

1. Other than jewelry, why is gold important to society?
2. What is the importance of chromite ore to our society?

What I can do

What do minerals do for the body? *(5 points)*

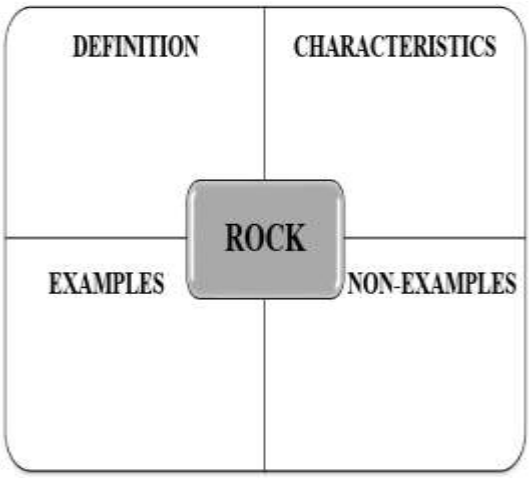
Assessment

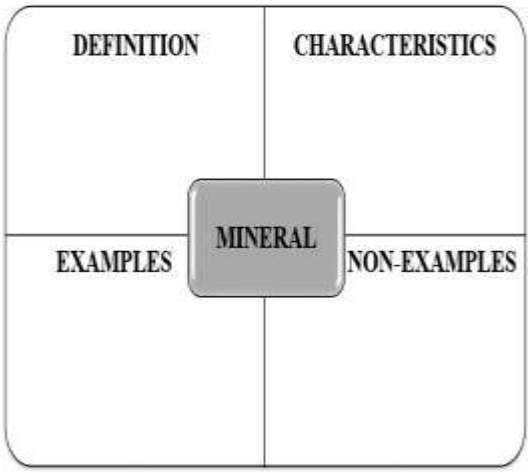
Directions: Match the minerals in Column A with their respective product in Column B. Write your answer on the space provided. *(10 points)*

Column A	Column B
_____1. Copper (element from mineral bornite)	A. car batteries
_____2. Diamond (mineral with chemical formula C)	B. cement
_____3. Graphite (mineral with chemical formula C)	C. drill bits
_____4. Gravel (rock fragments)	D. electrical wire
_____5. Iron (element from mineral magnetite)	E. glass
_____6. Lead (element from mineral galena)	F. concrete
_____7. Limestone (rock containing the mineral calcite)	G. pencil lead
_____8. Quartz sand (a mineral with formula SiO_2)	H. steel
_____9. Fluorite (the mineral form of calcium fluoride)	I. cosmetics
_____10. Talc (a clay mineral with a chemical formula 3MgO)	J. toothpaste

Additional Activities

Directions: Complete the four-square graphic organizer below by supplying the required information about rocks and minerals.





REFERENCES

"Common Minerals and Their Uses." Accessed June 5, 2020. https://www.chegg.com/homework-help/questions-and-answers/activity-1- common-minerals-uses-demonstrate-vital-nature-variety-minerals-use-ingredients-- q32487515.

"Minerals in Your House," 2015. http://www.mineralogy4kids.org/minerals-your-house.

Ogle, D. "Mineral Match Up." Accessed June 8, 2020. https://www.pinterest.ph/pin/469429961135061918/?nic_v1=1ak7k2yvpa5W2OiTJ3uovHrKqQXL1E68jdJFZLXn1lHjrEQJn9w1vIv42eNlhM/pDk.

Quinn, A. "Rock Cycle Interactive," 2007. https://www.learner.org/series/interactive-rock-cycle/.

ScienceSpot. "Rock Cycle Crossword Puzzle," 2017. https://www.tes.com/teaching-resource/rock-cycle-crossword-puzzle-11279813.

Fossil Fuels Formation and Ore Minerals
Quarter 1 – Week 3

What I Need to Know

This module was designed and written with you in mind. This worksheet contains activities that you will perform as an analogy for separating minerals in the mining industry. You may answer directly to this activity sheets and make sure to follow the directions stated in each part of the activity. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

The module comprises mainly of:

- Describe how ore minerals are found, mined, and processed for human use.
- Describe how fossil fuels are formed.

After going through this module, you are expected to:

1. Perform an analogy for separating minerals in the mining industry.
2. Execute and simulate how the way actual minerals are mined from the earth.
3. Demonstrate and discover the origin of fossil fuels.
4. Explore various aspects of fossil fuels by examining the various ranks of coal and the processes by which coal, oil, and natural gas form.

What I know

LET'S TRY! (PRE-ASSESSMENT)

Directions: How many of these items can you find around your house? Check the box as you discover each one!

MINERAL	USES	FACTS
BARITE	☐ Rubber hoses for dishwashers and washing machines ☐ Reflective paints ☐ Bowling balls, golf balls, tennis balls (adds density) ☐ Plastics	Barite has been mined in Bartow County for more than 100 years. Barite ore is usually massive, but can also come in beautiful crystals.
BAUXITE (ALUMINUM ORE)	☐ Aluminum foil ☐ Soda cans ☐ Kitchen utensils, pots and pans ☐ Metal window frames and screens	Georgia was the nation's first bauxite producer, and was first mined on what is now Barnsley Resort. The old bauxite pits are now part of the resort's sporting clay grounds.

FELDSPAR

- ☐ Dinnerware
- ☐ Window glass
- ☐ Ceramic sinks and toilets
- ☐ Floor tiles

Feldspar is one of the world's most abundant minerals, and when it weathers, in turns into another important Georgia mineral—kaolin.

MINERAL	USES	FACTS
GOLD	☐ Jewelry ☐ Electronics (computers, cell phones, etc.) ☐ Electronics in appliances ☐ Dentistry (fillings)	In 1829, Georgia experienced the first major gold in the country. Mining continued off and on until 1932, more than a hundred years later. And there's still gold out there—a specimen containing approximately seven ounces of gold was found by a prospector only 25 years ago and is now on display at Tellus along with 130 nuggets found with it.
GRAPHITE	☐ Pencil lead ☐ Lubricant for door locks ☐ Lightweight bicycle frames ☐ Batteries	Graphite is so soft you can write on paper with it—that's why it's used for pencils (remember, pencil "lead" is not made of lead; it's made of graphite).
HEMATITE AND GOETHITE (IRON ORES)	☐ Steel (appliances, screws and nails, tools) ☐ Pigments in grout ☐ Cosmetics	Georgia was a major iron producer until the end of the Civil War. A few mines opened after, but the last one closed in 1958. Goethite (pronounced gerr-tite) was the major iron ore mined in the area around the Museum.
KAOLIN	☐ Glossy paper ☐ Toothpaste ☐ Paint ☐ Cosmetics ☐ Rubber hoses for dishwashers and washing machines ☐ Concrete	Kaolin—mined in central Georgia—is the state's most valuable mineral product. It is even exported to China!

Photo Credit: https://betterlesson.com/lesson/641211/mining-for-ore

Day 1 & 2	How do we extract minerals?

Ore is a natural occurrence of rock or sediment which contains enough minerals with economically important elements, typically metals, which can be extracted from the deposit economically. The ores are extracted by mining for a profit from the earth; they are then refined (often by smelting) to extract the valuable elements.

At the end of this unit, you are expected to:

1. Describe how ore minerals are found, mined, and processed for human use. **[S11ES-Ic-d-8]**

What's In

The ore quality, and density of a rock or metal ore, as well as its occurrence type, can directly affect the ore mining costs. It is therefore necessary to weigh the extraction cost against the metal value contained in the rock to determine which ore can be processed and which ore is too low a grade to be worth mining.

What's New

The primary methods used to extract minerals from the ground are:

1. Underground mining
2. Surface (open pit) mining
3. Placer mining

The location and shape of the deposit, the strength of the rock, ore grade, mining costs, and the current market price of the commodity are some of the determining factors for selecting which mining method to use.

Higher-grade metallic ores found in veins deep under the Earth's surface can be profitably mined using underground methods, which tend to be more expensive. Large tabular-shaped ore bodies or ore bodies lying more than 1,000 feet (300 m) below the surface are generally mined underground as well. The rock is drilled and blasted, then moved to the surface by truck, belt conveyor, or elevator. Once at the surface, the material is sent to a mill to separate the ore from the waste rock.

Lower-grade metal ores found closer to the surface can be profitably mined using surface mining methods, which generally cost less than underground methods. Many industrial minerals are also mined this way, as these ores are usually low in value and were deposited at or near the Earth's surface. In a surface mine, the hard rock must be drilled and blasted, although some minerals are soft enough to mine without blasting.

Placer mining is used to recover valuable minerals from sediments in present-day river channels, beach sands, or ancient stream deposits. More than half of the world's titanium comes from placer mining of beach dunes and sands. In placer operations, the mined material is washed and sluiced to concentrate the heavier minerals.

What is It

In this activity, you will separate beads as an analogy for separating minerals in the mining industry.

Materials:

peanuts	tap water
raisins	two tall glasses or beakers
soda water	

Procedure:

1. Pour tap water into the first glass until it is about ¾ full.
2. Add a handful of the peanuts and raisin mixture to the water and note what happens.
3. Pour soda water into the second glass until it is about ¾ full.
4. Add a handful of the peanuts and raisin mixture to the soda water and note what happens.
5. Write down your observations.
6. Explain your observations. Use the following space to record and explain your observations.

Peanuts and raisins.

Separating peanuts and raisins

Looking down into the water filled beaker.

Photo Credit: https://www.womeninmining.org/wpcontent/uploads/2016/01/MineralsandtheProductsofMiningeditedemp7-7-14okaw.pdf

Guide Questions:

1. How did you sort the beads based on size?

2. How did you sort the beads based on shape?

3. How did you sort the beads based on density?

4. How did you sort the beads based on magnetic properties?

What's More

Materials:

chocolate chip cookies	toothpick
sticks	paperclip

Procedure:

1. Work with your parents or with your siblings.

2. Get your chocolate chip cookies. The cookie represents the land, and the chocolate chips are ore deposits.
3. Place a cookie on the first Mineral Deposit box and you have three minutes to mine using the mining tools one trial at a time, placing the ore in the Ore Deposit box.
4. Write down your observations in the Mining for Ore Investigation Sheet.

Trial 1	Tool Used: **Toothpick**
Mineral Deposit	Ore Deposit
Trial 2	Tool Used: **Sticks**
Mineral Deposit	Ore Deposit
Trial 3	Tool Used: **Paperclips**
Mineral Deposit	Ore Deposit

Guided Questions:

1. How effective were your tools at extracting the ore? Give specific examples.
2. How well were you able to return the land to its original state? Give a specific example.
3. How can you relate this to the way actual minerals are mined from the earth?

What I have Learned

Directions: Briefly answer the given questions.

1. Why does ore need to be extracted from rock?
2. How is ore extracted from rock?
3. Why does extracting ore from rock take a lot of energy?

What I can do

Everything we use on Earth that is not made of plants or animals is made of minerals. These minerals are our natural resources. They are mined so that we can have all of the products we're used to using. Even though over 99 percent of the Earth's surface has never been mined, it is important to remember that minerals exist in limited supply. We should be aware of what products they provide us with and use our mineral resources wisely.

Directions: As many as you can, list down some ways to preserve our mineral resources wisely.

Assessment

Directions: Circle the letter of the best answer.

1. What are solid, naturally occurring mineral deposits that can be extracted from the earth for an economic profit?
   A. Ores B. metals C. soils D. rocks
2. Before the extraction of ore, what factor must be weighed to determine whether it is beneficial to extract?
   A. the concentration factor
   B. the overall grade of the ore
   C. the size of the ore deposit
   D. the market value of the ore

3. Soluble metal compounds tend to be found in the _____, whereas insoluble metal compounds tend to be found in the _____.
   A. oceans; earth's crust
   B. salt beds; oceans
   C. rivers; oceans
   D. earth's crust; salt beds
4. Which of the following statement will a company is likely to mine?
   A. is rich in minerals
   B. contains gold
   C. does not contain gangue minerals
   D. contains minerals that can be economically extracted
5. Which of the following activities at a mining site will cause the least amount of harm to wildlife?
   A. dredging
   B. removing soil from an area that is to be surfaced mined.
   C. replacing the soil, with the lowest layers going in first.
   D. removing vegetation from an area that is to be surface mined.

Day 3 & 4 | Bread Fossils: Discover the Origin of Fossil Fuels

You may answer directly to this activity sheets and make sure to follow the directions stated in each part of the activity. Answer all questions the best that you can and write eligibly.

Hello, young readers! In today's lesson, we will learn about the formation of three main types of fossil fuels- petroleum, gas, and coal. So, keep on learning to find how fossil fuels are formed. I'm sure you'll enjoy it!

This worksheet contains activities that you will explore various aspects of fossil fuels by examining the different ranks of coal and the processes by which coal, oil, and natural gas form.

At the end of this unit, you are expected to:

1. Describe how fossil fuels are formed. [**S11ES-Id-10]**

What's In

If you have ever walked along the bottom of a cliff, you may have noticed that the rocks form layers. Different layers may have different colors or textures. They may be from bits of other rocks.

Rocks form layers like these over millions of years. As the layers build-up, the pressure on the bottom layers increases. The pressure on the rocks causes their temperatures to increase as well. Sometimes, rock layers form over the remains of plants and animals. Scientists call these remains organic matter. High pressures and temperatures can change organic matter into the three most important kinds of things: coal, oil, and natural gas.

Coal, oil, and natural gas are also called fossil fuels. Why do you think this is so?

What are fossil fuels? Fossils are the remains of creatures that lived long ago. So, fossils include organic matter buried beneath layers of rocks. Fuel is a source of energy. Without fossil fuels, most people could not drive their cars. They could not turn on their lights or heat their homes. It is because most of the energy needed to do these things comes from fossil fuels.

The energy in fossil fuels originally came from the Sun. Plants use the energy in sunlight to make their own food. The energy in plants passes to the animals that eat the plants. (You can learn about these processes in the lesson on Food Webs.) Some energy remains in plants and animals that die and become fossil fuels. Burning fossil fuels releases energy for humans to use.

What's New

The energy in fossil fuels originally came from the Sun. Plants use the energy in sunlight to make their own food. The energy in plants passes to the animals that eat the plants. (You can learn about these processes in the lesson Food Webs.) Some energy remains in plants and animals that die and become fossil fuels. Burning the fossil fuels releases energy for humans to use.

Fossil fuels are non-renewable energy sources that formed more than 300 million years ago during the Carboniferous Period - long before dinosaurs roamed the Earth. Fossil fuels are made up of plant and animal matter. When plants and animals died, their bodies decomposed and were buried under layers of earth. Millions of years later, we have the three forms of fossil fuel: oil, natural gas, and coal.

What is It

PART 1 - Bread Fossils: Discover the Origin of Fossil Fuels

What do fossils have to do with energy? Conduct these bread fossils experiment to discover the origin of fossil fuels.

Materials:

3 slices of bread (one slice each of white, wheat, and rye)
Heavy books
Gummy candy
Magnifying lens
Paper towels
Clear drinking straws

Procedure:

1. Carefully pull the crust from the three slices of bread.
2. Place a piece of white bread on top of the paper towel. Put two or three gummy candies on top of the white bread.
3. Place a piece of rye bread on top of the white bread layer. Put two or three gummy candies on top of the rye bread.
4. Place a piece of wheat bread on top of the rye bread layer.
5. Fold the paper towel to cover your bread fossil.
6. Place two textbooks or a heavy object found in the room on top of the bread. Place your bread fossil with the pressure source in a secure area of your home.
7. Write down your observations on the table and answer the guided questions below.

Day 1 Written Description	Day 1 Drawing
Day 2 Written Description	Day 2 Drawing

Guided Questions:

1. What piece of bread looks like on the sandy floor of the ocean?
2. What layer of bread could we use to represent the sediments?
3. What does the last layer of bread represent?
4. What was used in your experiment to put pressure on the "rock layers" of the "bread fossil?"

What's More

PART 2 -Bread Fossils: Discover the Origin of Fossil Fuels

Explore various aspects of fossil fuels by examining the different ranks of coal and the processes by which coal, oil, and natural gas form.

Materials:

Your previous output Bread Fossils clear straw plastic knife

Procedure:

1. Unfold your "bread fossil." Place the straw in your fossil into the middle of the bread fossil to "extract" a core sample.
2. Remove the core sample and observe the layers through the straw. Record your observations in words and draw a picture on the chart for day 2 on page 2 of this packet.
3. Take your knife and cut the bread fossil in half. Look at the layers of your fossil. Record your observations.
4. Pick up your bread fossil and try to separate the layers of the bread. Try to extract the fish.

Guided Questions:

1. What do you see in the straw?
2. What do you see when you cut into the bread fossil?
3. Are there any similarities and differences between the straw sample and the bread fossil?
4. Describe how the core sample of the "bread fossil" looked.
5. Why do you think the layers are difficult to separate?
6. Can you identify the mold and the cast of the gummy fish?

What I have Learned

Observe the colored residue of the gummy fish in the bread fossil. Compare it to the remains of the plants and animals that seep into the rock. The residue left by the gummy fish represents oil deposits left behind by dead plants and animals. Over millions of years, these remains are pressurized to become an oil and natural gas deposits. As we journey back in time, let's think about how we can recreate the historical formation of fossils. What eventually happens to sea animals and plants when they die? (They fall to the ocean floor.) As the plants and animals lay

lifeless, wind and currents deposit sediments on top of the dead. As these layers increase, the pressure also increases, creating fossils and fossil fuels.

Guided Questions:

1. What has changed about our "bread fossils?" What happened to the layers?
2. Did you ever hear that oil and natural gas are fossil fuels?
3. Do you think oil and natural gas can be produce from old fossils?
4. How long do you think it takes fossil fuel to form?

What I can do

Problem: Pollution due to Overuse of Fossil Fuels

We depend on non-renewable sources of energy such as fossil fuels and are responsible for most of the CO2 emitted. We agree that renewable energy sources are the answer to future energy supplies. We understand that it will take time and resources to convert to green energy, but it is necessary for a sustainable future.

Think of possible solutions to the world's overuse of fossil fuels and enumerate it on the space provided.

_____

_____

_____

Assessment

Directions: Read and analyze each question carefully, then circle the letter of the BEST answer.

1. Fossil fuels are classified as non-renewable because they _____.
   A. require expensive equipment
   B. take millions of years to replace
   C. can be found all over
   D. provide all the energy we use
2. Which of the following is NOT a type of fossil fuel?
   A. coal
   B. light
   C. petroleum
   D. natural gas
3. What are fossil fuels?
   A. dinosaur bones
   B. resources made from once-living organism
   C. combination of different rocks
   D. gas stations
4. Why is it important to save energy in our daily lives?
   A. We need to burn more fossil fuels.
   B. So that other people can waste energy.
   C. We save electricity because it's easier to see in the dark.
   D. We need to protect our environment for the future.
5. Which statement is true?
   A. Coal has energy stored in it because it is black.
   B. Fossil fuels have energy stored in them because the animals that became fossils ran around a lot.
   C. Coal has energy stored in it because it is warm inside the Earth.
   D. The plants that formed coal got their energy from the Sun.

Additional Activities

Directions: Work with your parents/ guardian to determine how much your family depends on fossil fuels. A typical household will likely consider four main categories of usage:

- **Electricity:** Do you use electricity generated by burning coal, oil, or natural gas?
- **Heating:** Do you heat your home using appliances that burn coal, oil, or natural gas? (Some heating appliances are powered by electricity.)

Instruct the members of your family to keep track of their activities for one week. Record how long you spend doing each activity that depends on fossil fuels. Try to measure how much fossil fuel is necessary for each activity. For example:

✓ How many kilowatt-hours of electricity is needed to run your television for 30 minutes? (The amount of fuel used to generate one kilowatt-hour of electricity equals the heat rate of the generator divided by the heat content of the fuel. You may not be obtain this information; regardless, the more electricity you use, the more fossil fuel is burned.)

REFERENCES

Aquilla ISD. Envi. Systems Instructional Packet April 20-24. (2020) https://4.files.edl.io/9b4f/04/20/20/173825-72ef7691-92a4-498b-a915-35682e7264e5.pdf

Australian Broadcasting Corporation ABC. Teacher Resource. Fossil Fuels. Retrieved from http://www.abc.net.au/btn/resources/teacher/episode/20150616-fossilfuels.pdf

Australian Institute of Geoscientists. Retrieved from https://www.aig.org.au/tag/mining/

Better Lesson. Mining for Ore. Retrieved from https://betterlesson.com/lesson/641211/mining-for-ore

Doc Player. The Formation of Fossil Fuels. Retrieved from https://docplayer.net/21239220-The-formation-of-fossil-fuels.html

Integrate Interdisciplinary teaching about Earth for a Sustainable Future. Unit 4: Fossil Fuel Formation. Retrieved from https://serc.carleton.edu/integrate/teaching_materials/change_inthe_air/activity4.html

National Energy Education Development Project. Fossil Fuels to Products. Retrieved From http://www.switchenergyproject.com/education/CurriculaPDFs/SwitchCurricula-Secondary-Oil/SwitchCurricula-Secondary-FossilFuelsToProducts.pdf

Nerdy Caterpillar. Retrieved from https://www.nerdycaterpillar.com/how-are-fossil-fuels-formed/

Paper and Pencil Activity, Fossil Fuel Use Retrieved from https://www.womeninmining.org/wpcontent/uploads/2016/01/MineralsandtheProductsofMiningeditedemp7-7-14okaw.pdf

Raymundo, Roberto (2019). Angelo King Institute of De La Salle University. Retrieved from https://www.dlsu.edu.ph/wp-content/uploads/2019/03/2015-25.pdf

Siyabula Technology Powered Learning. Extracting Ores. Retrieved from https://www.siyavula.com/read/science/grade-9/mining-of-mineral-resources/25-mining-of-mineral-resources?id=toc-id-3

Tellus Science Museum in Facebook. Retrieved from. https://web.facebook.com/tellusmuseum/photos/pcb.10157222409847883/10157222405777883/?type=3&theater

The National Association of Geoscience Teachers. Ore Exploration Game. Retrieved From https://nagt.org/nagt/teaching_resources/teachingmaterials/58649.html

Women in Mining Education Foundation. Retrieved from https://www.womeninmining.org/wpcontent/uploads/2016/01/MineralsandtheProductsofMiningeditedemp7-7-14okaw.pd

Energy and Water Resource
Quarter 1 – Week 4

What I Need to Know

This module was designed and written with you in mind. It is here to help you understand how heat is generated from inside the Earth and how we tap this heat as a source of energy for human use. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order you read them can be changed to correspond with the textbook you are now using.

The module comprises mainly of:

- Explain how heat from inside the Earth (geothermal) and from flowing water (hydroelectric) is tapped as a source of energy for human use. **(S11ES-Ig-11)**
- Identify water resources on Earth. **(SLES-If-G-15)**

After going through this module, you are expected to:

1. Explain how heat inside the Earth is tapped as an energy source for human use.
2. Explain how hydroelectricity is generated.
3. Identify the various water sources on Earth.

What I know

LET'S TRY! (PRE-ASSESSMENT)

Directions: Fill in the K-W-L Chart below to assess your prior knowledge and understanding of the topic of how heat inside the Earth and flowing water are tapped as energy sources, and what are available water resources for human use.

What do I Know?	What do I Wonder?	What did I Learn?

Day 1 — Energy from Heat Underneath

At the end of this unit, you are expected to:

- Explain how heat inside the Earth is tapped as an energy source for human use.
  S11ES-Ig-11

What's In

What do we call the heat generated from inside the Earth? How do we tap this heat as a source of energy for human use?

The heat from inside the Earth is called **geothermal energy**. It is considered a renewable resource because the heat from the core of the Earth is limitless and it continuously flows outward. Heat, in the form of magma, when it reaches the surface of the Earth will be called lava. When magma flows outward and remains beneath the crust, it heats nearby rocks and water. When water is heated, it produces steam or hot water trapped in permeable and porous rocks under a layer of impermeable rock, and a geothermal reservoir can form. This hot geothermal water can materialize on the Earth's surface as hot springs or geysers.

Geothermal energy can be used as direct heating systems, electricity generators, and as heat pumps. Some of these applications use the heat near the surface or require drilling miles into the Earth.

1. Direct use of geothermal energy includes the use of hot mineral springs for bathing, cooking, and heating. Energy from hot water near the earth's surface can be piped directly into buildings and industries for heating purposes.

2. Geothermal power plants use water (hydro) and heat (thermal) or hydrothermal resources with temperatures between 300 to 700 °F (149 to 371°C) and may come from either dry steam wells or hot water wells one to two miles deep. Hot water under high pressure from deep underground is pumped through a well. As this water approaches the surface with lower pressure, it turns into steam. This steam causes the turbine to spin, which connects to a generator that produces electricity. The steam cools off in a cooling tower and condenses back to water before pumping back into the Earth to begin the process again. Geothermal power plants have three basic types; dry steam plants, flash steam, and binary power plants. Among the three, the commonly used are Flash steam plants.

3. Geothermal heat pumps use the almost constant temperature from the upper 10 feet of the Earth's surface to heat and cool buildings. Water or a refrigerant moves through a loop of pipes. Throughout winter or during cold weather, the water or refrigerant heats up as it travels through the pipelines that are buried underneath the ground. The warmed water or refrigerant then transfers energy (heat form) into the building once it gets back above the ground. The water or refrigerant cools down after its heat is transferred. It is pumped back underground and heats up once more **and,** the process then starts again. The system can run in reverse during hot days. The water or refrigerant cools the building and then is pumped underground where captured heat transfers to the ground around the pipes.

What's New

Hydropower Plants

The main components of a hydropower plant are a dam, intake, turbine, generator, transformer, power lines, and outflow. Most hydropower plants rely on a dam that holds back water, creating a large reservoir. Large dams are vital for large-scale hydropower, but dams of all sizes are also used for flood control, water storage, and irrigation throughout the world. Gates on the dam open, and gravity pulls the intake water through the penstock. It is a pipeline that leads to the turbine. Water builds up pressure as it flows through this pipe. The water strikes and turns the large blades of a turbine. It is attached to a generator above it by way of a shaft. As the turbine blades turn, so do a series of magnets inside a generator. Giant magnets

rotate past copper coils, producing alternating current (AC). A transformer inside the powerhouse takes the AC and converts it to a higher-voltage current that carries high-tension power lines. The used water is carried through pipelines called tailraces, and this outflow re-enters the river downstream.

What is It

Directions: Study the diagram of a geothermal power plant above. Draw arrows to follow the path of hot water or steam to complete the process of generating electricity from geothermal energy.

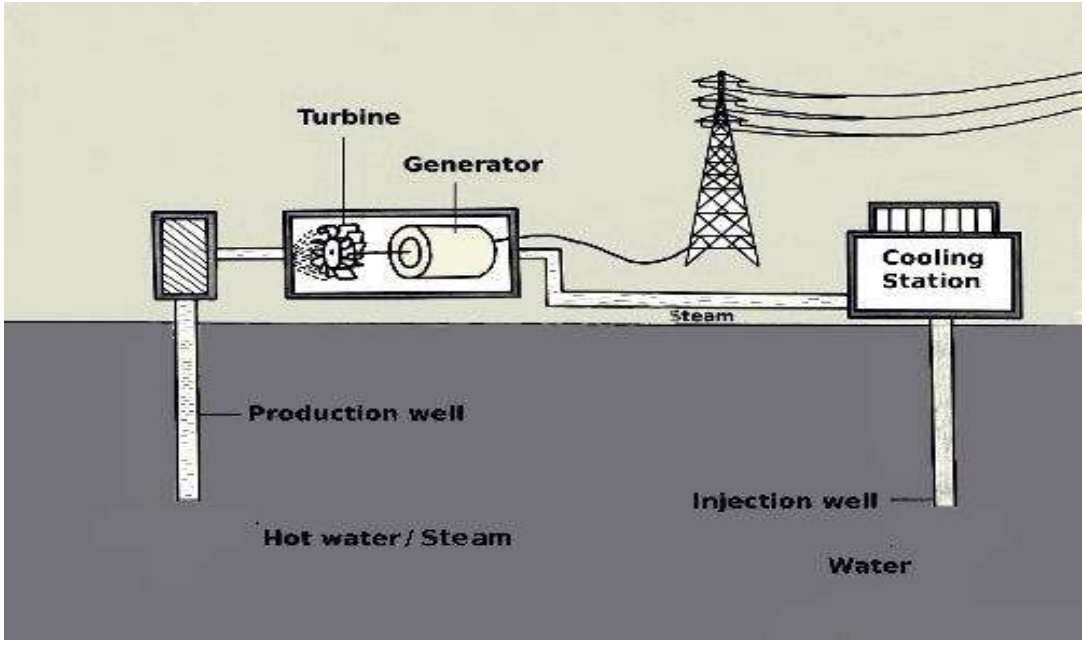


Figure 3: Diagram of a Geothermal Plant

Guided Questions

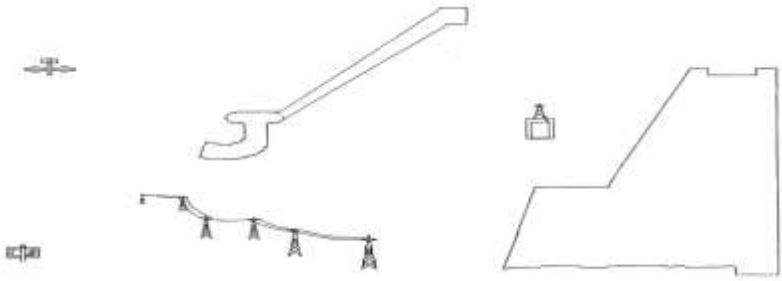
1. What do you call the part of the Earth where hot water or steam is located?
2. What passes through the production well?
3. What makes a turbine spin?
4. What happens in a cooling station?
5. What passes through an injection well?
6. What happens to the water after it pumps back below the surface?

What's More

Part A.

Directions: Cut out each of the six parts and glue them to the background page. You need to add the water; indicate the water in your diagram by coloring it in.

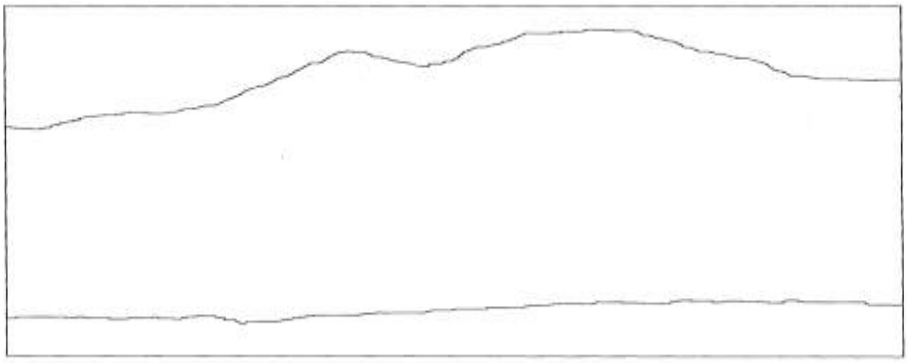
ELEMENTS OF A HYDROELECTRIC POWER STATION



Part B.

Directions: Glue the six hydroelectric power station parts onto the background. The river bed and backdrop of hills are sketched for you. Don't forget to add the water and indicate where it is by coloring it in. Wherever possible, label the diagram and provide additional descriptive notes to explain your understanding of the function of the parts and the station.

HYDROELECTRIC POWER STATION SCHEMATIC DIAGRAM



What I have Learned

Directions: Answer the following questions:

1. Explain how forces can affect movement.
2. How **can** potential energy **transform** into kinetic energy? Relate the transformation of energy to the intended uses of the energy.
3. Explain how energy can be stored in water in a dam. Represent the transfer and transformation of energy using models or conceptual flow diagrams.
4. Explain the conversion of energy into other forms and the effects of that conversion. Use the concepts of power and efficiency to explain physical situations of energy transfer and transformation.

What I can do

List down countries with active geothermal power plants and give additional information by completing the table below.

Countries with geothermal power plants	Continent	Number of active volcanoes

Assessment

Directions: Create your model on how geothermal energy is tapped for human use. The models will serve as a learning reinforcement of the knowledge you have acquired. (Place your answer in a clear bond paper)

Rubrics for assessing models:

1. Completeness of model in detailing how geothermal energy is tapped for human use - 40%
2. Conciseness and clarity of presentation/explanation of the model - 30%
3. Creativity and resourcefulness in creating the model - 30%

Day 2 Water Resources

Water resources of the Philippines include inland freshwater (rivers, lakes, and groundwater) and marine (bay, coastal, and oceanic waters). Overall, there is sufficient water but not enough in highly populated areas, especially during the dry season.

At the end of this unit, you are expected to:

1. Identify water resources on Earth. **(SLES-If-G-15)**

What's In

Water is a simple compound that covers about 71% of the earth's surface. It is one of the most significant substances on our planet and one of the main reasons why Earth is habitable. Without water, life could probably not even exist on Earth. The abundance of this substance in our planet is evident from its color when viewed from outer space. But did you know that only 3% of the earth's water is fresh while the remaining 97% resides in the ocean? From this freshwater percentage, 69% exist in glaciers, 30% are underground water and less than 1% are located in lakes, rivers, and swamps. About 10% of the earth's freshwater – neither frozen nor underground, is found in the atmosphere and the movement of water from the atmosphere to the ground is best explained by the water cycle diagram.

I	U	R	U	N	O	F	F	J	K	G	N	L	E	T	Q	O
A	O	R	J	I	P	H	B	E	F	L	Z	L	S	O	L	H
P	W	G	L	O	I	R	S	D	N	A	P	E	W	I	H	H
F	R	E	S	H	W	A	T	E	R	C	J	W	A	W	R	E
D	E	H	S	R	E	T	A	W	R	I	Y	R	M	S	E	G
M	U	Z	J	U	F	S	D	F	N	E	L	D	P	E	T	N
K	L	V	I	W	M	P	C	A	X	R	D	V	S	Z	A	I
C	A	Q	U	I	F	E	R	S	W	S	Q	L	E	S	W	R
W	B	W	Z	V	A	A	I	L	J	Q	K	Z	R	W	D	P
B	C	L	J	K	Y	J	S	A	D	N	A	E	C	O	N	S
I	N	W	F	Z	F	K	F	K	F	S	T	R	M	A	U	L
V	M	T	Q	L	W	W	N	E	I	S	Q	A	B	M	O	A
U	H	N	L	P	O	N	D	S	Q	G	E	Y	S	E	R	R
V	G	D	Q	J	N	Q	N	D	R	Q	O	Z	J	H	G	U
K	R	P	T	S	G	R	E	B	E	C	I	E	V	Y	G	T
D	E	A	H	J	L	Z	U	K	L	G	T	N	I	J	P	A
Z	E	M	E	T	S	Y	S	R	E	V	I	R	F	I	X	N

What's New

The world's **water** exists naturally in different forms and locations: in the air, on the surface, below the ground, and in the oceans. Just 2.5% of the **Earth's water** is freshwater, and most are frozen in glaciers and ice sheets. About 96% of all liquid freshwater can be found underground. Our main sources of water for drinking, washing, agriculture, and industry are surface water, groundwater, and collected rainwater, all of which are dependent on rain and snow falling on the Earth's surface.

What is It

Directions: This word search will familiarize you with the different water resources on Earth. Locate the given words within the grid by encircling it. Words may run horizontally, vertically, diagonally, and even backward.

WORD POOL

Aquifer	Fresh Water	Geyser	Glaciers	Groundwater
Icebergs	Lakes	Natural Spring	Ocean	Ponds
River System	Run-Off	Swamps	Water Shed	Well

What's More

Guided Questions:

1. How much water is there on earth and how much is available for humans?
2. Identify five (5) different sources of water for urban supply. Describe their characteristics, and explain how water is abstracted from them.

1	2	3	4	5

What I have Learned

In today's session, I was able to know and understand…

What I can do

Directions: List down how do human activities affect the quality and availability of water on Earth?

1._____

2._____

3. _____

4. _____

5. _____

Assessment

Directions: Write **"FACT"** if the statement is true, and "**BLUFF"** if the statement is wrong.

_____1. Groundwater comes from underground lakes and rivers.

_____2. Abandoned water wells are better left alone.

_____3. Once groundwater is contaminated, it cannot be restored.

_____4. Surface water and groundwater are two separate waters.

_____5. Contaminants can quickly pollute groundwater.

Additional Activities

Directions: Give at least 5 advantages of water resources in your community and cite some examples. Place your answer in a clean sheet of paper. (25 points)

REFERENCES

Darling Kindersley Limited (20100 Retrieved from
https://www.greatschools.org/library/cms/15/25815.pdf
iSLCollective Website Retrieved from
https://en.islcollective.com/english-esl-worksheets/skill/reading/water-cycle-diagram/41121
Teach Engineering STEM Curriculum for K to 12. Retrieved from
https://www.teachengineering.org/activities/view/cub_environ_lesson09_activity3
Wikiversity Website. Power Generation/Hydro Power. Retrieved from
https://en.wikiversity.org/wiki/Power_Generation/Hydro_Power
Blodgett, Leslie, and Kara Slack, 2009. 'Basics of Geothermal Energy Production and Use'.
https://geothermalcommunities.eu/assets/elearning/7.15.geo_101.pdf
'Geothermal Basics', 2008. https://ei.lehigh.edu/learners/energy/readings/geothermal.pdf
Global Climate Change. 'Geothermal Energy', 2017.
https://archive.epa.gov/climatechange/kids/solutions/technologies/geothermal.html
Jeon, Jun-Seo & Lee, Seung-Rae & Minjun, Kim. (2018). A modified mathematical model for spiral coil-type horizontal ground heat exchangers. Energy. 152. 10.1016/j.energy.2018.04.007.

Activities Affecting Water and Soil Resources Quarter 1 – Week 5:

What I Need to Know

This module was designed and written for you to help you learn more about energy and water resources. The scope of this module permits it to use in many different learning situations. The language used recognizes the diverse vocabulary

level of students. The lessons are arranged to follow the standard sequence of the course:
The module is divided into two lessons, namely:

Lesson 1 – Activities Affecting Water Quality and Availability
Lesson 2 – Activities Affecting Quality and Quantity of Soil

What I know

LET'S TRY! (PRE-ASSESSMENT)
Directions: Fill in the K-W-H-L Chart below to assess your prior knowledge and understanding of the topic activities affecting the quality and availability of water for human use, and activities affecting the quality and quantity of soil.

K	W	H	L
What do I know?	**What do I want to find out?**	**How can I found out what I learn?**	**What did I learn?**

Day 1 Activities Affecting Water Quality and Availability

Water comprises 71% of our planet's mass. Even our body is made up of water. We do some of our daily chores with water. Soil, on the other hand, is known as the "skin of the Earth". We live with the soil beneath our feet. We gather our food from plants grown from the soil. Both water and soil are essential in our daily living that is why they are included in our Earth's resources. With the rapid growth of population and industrialization, the needs for these resources are increasing dramatically. This module is made to understand how different activities affect our water and soil resources.

At the end of this unit, you are expected to:

- Explain how different activities affect the quality and availability of water for human use. **S11ES-Ig-16**

What's In

Our planet is made up of resources that support the life inside it. We have previously learned about Energy resources which is one of the resources humans have learned to utilize. In this module, we will look into other resources that we consider essential to our existence.

What's New

Water resources can classify as groundwater and surface water resources that can be harvested for human consumption. When talking about water consumption, the terms water quality and availability always follows. Water quality is an indicator of how water is suitable for a particular use based on selected physical, chemical,

and biological characteristics. The availability of quality water is a key determinant for the survival of humans, plants, and animals. The decline in water quality reduces the available water supply to serve different uses.

Water resources face serious threats caused primarily by human activity like urban and industrial development such as farming, mining, animal-feeding operations, and extracting too much water, both from surface and groundwater. Urban development includes the converting of land into residential areas. Industrial development aims to develop a particular area to be part of an industrial or manufacturing enterprise. Farming defines as the activity of cultivating land to grow crops and raise livestock. Mining is a process of extracting useful mineral resources from the surface of the Earth. Animal-feeding operations are operations in agriculture where different animals are kept and raised in confined situations. Too much extraction of water – either surface or groundwater, could lead to problems regarding the water supply.

Each of these activities has its specific impact on ecosystems which will eventually impact water resources. To give an example of the effects of these activities on water quality, consider nitrogen and phosphorus fertilizers that are applied to crops and lawns. These plant nutrients can be dissolved easily in rainwater or snowmelt runoff. Nutrients that weren't absorbed by plants are carried to streams and lakes. It encourages abundant growth of algae, which leads to low oxygen in the water and the possibility of fish kills.

What is It

Directions: Do the following activities and answer the questions that follow.
Note: Before doing each activity, please collect a sample of clean water from the source in your house. During the activity, collect a sample of water after it is used. After the activity, place each sample side by side (clean and used water). Don't forget to label the containers containing your samples (e.g., sample 1 clean / sample 1 used).

1. Wash fruits or vegetables.
2. Take a bath.
3. Wash your hands with soap.
4. Wash the dishes.
5. Walk barefoot and then wash your feet.

Guided Questions:

1. What do you observe in your sample of clean water?
2. Did you spot similarities and differences with your samples of used water?
   - 2a. Cite some similarities that you observed.
   - 2b. Cite some differences that you observed.
3. What do you think happened to the water? What caused it?
4. What will happen to water quality if contaminants are eroded to streams and rivers?
5. What will happen to our available water supply if there is a continued decrease in water quality?

What's More

Directions: Provide possible scenarios of how water quality and availability will be compromised with the following human activities.

Table 1. Activities and Scenarios

Activities Affecting Water Quality & Availability	Possible Scenarios
Urban development	
Industrial development	
Farming	
Mining	
Animal-feeding operations	
Extracting too much water	

What I have Learned

I have learned that

_____

What I can do

How do you think an industrial plant built near a stream channel between a dam and a treatment plant could affect the quality of water? How does this affect the consumers of water from the treatment plant?

Assessment

Directions: Identify which human activity affects water quality and availability is being described. Write your answers on the space provided.

_____1. Leaching of minerals from an open site to the nearest stream or river.
_____2. Expansion of an unpopulated area into residential area.
_____3. Industries populating in an area
_____4. Cultivating land to grow crops and livestock
_____5. Use of deep well for irrigating crops and personal consumption

Day 2 — Activities Affecting Quality and Quantity of Soil

Soil, known as the skin of the earth, forms at the surfaces of the land and comprises the complex mixtures of minerals, water, air, organic matter, and the remains of countless decaying organisms. Soils differ according to how different factors interact to form them. These factors include climate, organisms, relief, parent material, and time.

At the end of this unit, you are expected to:

1. Identify human activities that affect the quality and quantity of soil. **S11ES-Ig-17**

What's In

Our planet is made up of resources that support the life inside it. We have previously learned about energy resources which is one of the resources humans have learned to utilize. In this module, we will look into another resource that we consider as essential to our existence namely, soil resources.

What's New

Soils have properties that help us describe and manage them. These are soil texture, soil structure, soil color, and soil profile. Soil texture depends on three particles namely sand, silt and clay. Each soil texture corresponds to specific percentages of the three. Soil structure refers to how soil particles is arranged into small clumps, called peds or aggregates. Structure correlates to the pore space in the soil that influences root growth and air and water movement. Soil color depends on what kind of minerals are present in that specific soil. This also tells us how soil behaves – if it drains well or is often wet. The soil profile is the term used to describe the different layers or horizons of soil.

Soil is as important as other resources. It provides nutrients for plant life and houses different kinds of microorganisms. It is considered a renewable resource because it is constantly being formed and destroyed, mainly by the process of weathering and erosion. As a recall, weathering is the breaking down of rocks and erosion is the movement of weathered rocks.

Soil erosion, especially when human-induced, is not always favored because it could lead to loss of nutrients and desertification. Agricultural depletion, overgrazing of animals, deforestation, mining operations, development and expansion, waste disposal, and recreational activities are some human activities that could speed up soil erosion.

What is It

Directions: Do the following activities and answer the guide questions that follow. Accomplish the table below.

1. Go to your front yard or backyard. Walk at least 10 steps from your starting point and observe how soil particles are moved with your steps.
2. Still in your front yard or backyard, run in circles as fast as you can. Observe how soil particles are moved with your steps.
3. Still in your front yard or backyard, run in place as fast as you can. Observe how soil particles are moved with your steps.

Table 2: Observation table

Procedure	Observation
1	
2	
3	

Guided Questions:

1. Describe the soil in your front yard or backyard.
2. While doing the activity, what changes have you noticed?
3. What do you think happened to the soil when you walk and run?
4. What will happen to the soil if you are to do the activity again and again?

5. Do you think plants and microorganisms could still flourish in that area if it is walked on or run by every day?

What's More

Directions: In a separate sheet of paper, discuss how the following activities could lead to soil erosion.

1. Agricultural depletion
2. Overgrazing animals
3. Deforestation
4. Mining
5. Urban development and expansion
6. Recreational activities

What I have Learned

In today's session, I was able to know and understand _____

What I can do

In an essay, explain how frequent hiking in a mountain affects soil's ability to support plant life.

Assessment

Directions: Match the human activities in column A to its examples in column B. Write your answer in the space provided.

Column A	**Column B**
_____1. Recreational Activities	a. farming
_____2. Development and Expansion	b. hiking
_____3. Mining	c. illegal logging
_____4. Deforestation	d. free-roaming farm animals
_____5. Overgrazing animals	e. building subdivisions
_____6. Agricultural depletion	f. throwing garbage on soils
	g. mountaintop removal.

Additional Activities

Directions: List down examples (of local places you know of) where any of the above-mentioned human activities have been observed or still being observed. Use the table below.

Table 2. Human Activities and Places

Human Activities	Places I Know

Reference/s
"Animal Feeding Operations". National Pollutant Discharge Elimination System (NPDES), June 10, 2020, accessed June 11, 2020. https://www.epa.gov/npdes/animal-feeding-operations-afos
Gail E. Cordy, "A Primer on Water Quality", USGS Science for a Changing World, February 18 2014, accessed June 9, 2020, https://pubs.usgs.gov/fs/fs-027-01/#:~:text=How do human activities affect,the quality of natural waters.
John Lawrence Mero, "Mining", Encyclopaedia Britannica, April 25, 2017, accessed June 9, 2020, https://www.britannica.com/technology/mining
Reference. 'Why Water Quality is Important?'. https://www.reference.com/science/water-quality-important-57b4c2bb03702ad1
"Soil Basics". Soil Science. https://www.soils4teachers.org/soil-basics
"Soil Resources". Cengage, May 28, 2020. https://www.encyclopedia.com/environment/energy-government-and-defense-magazines/soil-resources#:~:text= Soil Resources 1,mainly by erosion... More
"Water Quality & Water Supply: Definition & Purpose." Study.com. September 8, 2015. https://study.com/academy/lesson/water-quality-water-supply-definition-purpose.html.
"Water Resources". Green Facts, January 15, 2008. http://www.greenfacts.org/en/water-resources/

Quarter 1 – Week 6: Soil Conservation and Protection, Waste Generation and Classification

What I Need to Know

This module was designed and written with you in mind. It intends to help you learn facts, concepts, and principles on soil conservation and protection, waste generation, and classification. The scope of this module permits uses in many different learning situations. The language used recognizes the diverse vocabulary levels of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into two learning competencies, namely:

- Learning Competency 1 – Give ways of conserving and protecting the soil for future generations
- Learning Competency 2 – Describe how people generate different types of waste (solid, liquid, and gaseous) as they make use of various materials and resources)

After going through this module, you are expected to:

1. Demonstrate erosion and determine its impact on bare soil.
2. Identify, describe, and compare with each other the methods of controlling soil erosion.
3. Classify different wastes generated through various human activities.
4. Describe how people generate wastes in their daily activities.

What I know

LET'S TRY! (PRE-ASSESSMENT)

Directions: Read each statement in the anticipation-reaction chart below. Put a check under column (1) if you AGREE with the statement or under column (2) if OTHERWISE.

BEFORE THE LESSON (ANTICIPATION)		STATEMENT	AFTER THE LESSON (REACTION)	
AGREE (1)	DISAGREE (2)		AGREE (3)	DISAGREE (4)
		1. Raindrop or splash erosion transports soil over short distances.		

		2. In strip-cropping, groundcover plants such as grasses are planted between fields of crops.		
		3. Terracing involves planting crops in curving rows to follow the shape or form of a hill.		
		4. Old cellphone battery is a hazardous waste that can be buried underground.		
		5. Sulfur dioxide and particulate matter are forms of household waste.		

Day 1 Methods of Controlling Soil Erosion

Soil conservation primarily depends on methods designed to hold soil in place and keep it fertile with nutrients. Different soil erosion control methods include mulching, crop rotation, conservation tillage, contour plowing or farming, terracing, strip-cropping, and windbreaks.

At the end of this unit, you are expected to:

1. give ways of conserving and protecting the soil for future generations. **(S11ES-IH-j-18)**

What's In

Splash erosion represents the first stage in the soil erosion process facilitated by water through rainfall. Following splash erosion, other types of water erosion take place to bring a massive movement of soil. These include **sheet erosion**, **rill erosion**, **gully erosion,** and **stream bank** erosion. These four are distinguished from each other based on the land topography of occurrence (whether the land is flat or sloping), the driving force (rainfall, rainstorm, or flood), and the amount of soil displaced.

What is It

Methods of Controlling Soil Erosion

- **Contour plowing/farming** - Tilling sloped land along lines of consistent elevation to conserve rainwater and to reduce soil losses from surface erosion.
- **Crop rotation** - Rotating cultivation of different crops in a field across different growing seasons; yields can change every season or after a set number of seasons.
- **Mulching** - Covering the soil with *mulches*, such as bark, wood chips, leaves, and other organic material, to preserve moisture and improve the condition of the soil.
- **No-till planting** – Planting seeds in the ground without first tilling (or plowing) the soil; dead plants from the previous crop remain on the ground, with their roots holding the soil.
- **Strip cropping** – Planting groundcover plants such as grasses in strips between fields of crops; strips of groundcover soak up rainwater and slow down runoff

- **Terracing** – Building step-like **terraces** onto the slopes of hills and mountains, preventing runoff from rushing downhill
- **Windbreaks –** Planting rows of trees and shrubs between fields to reduce the force of the wind, thus, minimizing wind erosion

What's More

Directions: Fill out Table 1 based on concepts presented in the previous section.

Table 1. Soil Erosion Control Methods (7 pts)

Method	Description
1. Mulching	
2. Conservation tillage	
3. Crop rotation	
4. Contour plowing/farming	
5. Terracing	
6. Strip-cropping	
2. Windbreaks	

Guided Questions:

1. How does each method of controlling soil erosion protect and conserve the soil? (2 pts)
2. Compare the methods of controlling soil erosion with each other, based on: (a) cost-effectiveness and (b) ease of application. (6 pts)
3. Distinguish between: (a) Crop rotation and Strip-cropping and (b) Contour plowing/farming and Terracing. (4 pts)
4. Which soil erosion control method do you think is MOST effective? Why? (2 pts)
5. Which soil erosion control method do you think is LEAST effective? Why? (2 pts)

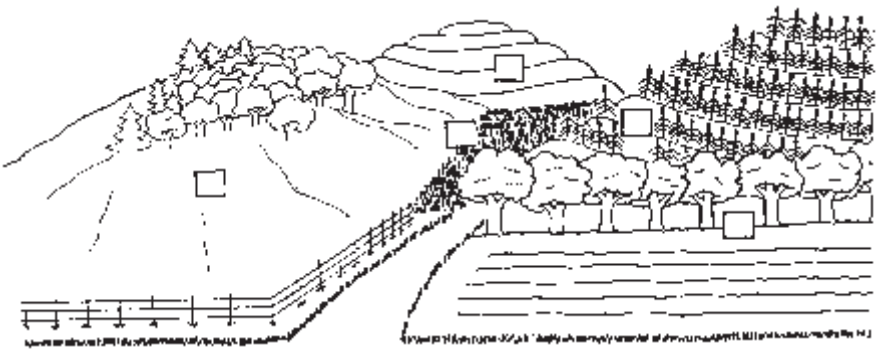
What I have Learned

1. What are the methods of controlling soil erosion? (3 pts)

2. How are these methods distinguished from each other? (7 pts)

What I can do

Directions: Analyze the image below and answer the succeeding activities below.



Source: https://education.hcswcd.org/uploads/1/5/4/8/15484824/dirt_secrets_in_the_soil_2.pdf

A. Match and write the number of soil erosion control methods in the CORRECT box in the image. (5 pts)

1. **Contour plowing/farming:** plant crops around the curve of a hill rather than up and down the hill
2. **Terracing:** wide ridges that go around a hill to prevent water from rushing down the hill too fast
3. **Forest and Grass Areas:** keep steep hillsides in trees or grass rather than clear for cropland
4. **Grassed waterways:** plant grass and do not plow low areas in a field where water usually runs
5. **Windbreak:** rows of trees planted to slow down the wind and prevent soils from

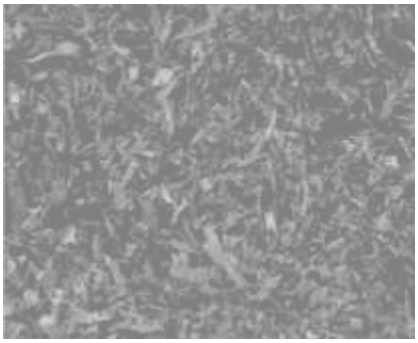
B. You have advised an inexperienced farmer to practice strip-cropping, but the farmer still wants to plant all the land in wheat to grow as much as possible. What reasons would you raise to BEST convince the farmer to follow your advice? (5 pts)

C. Design a method of soil conservation (maybe an entirely new method, a modification, or a combination of the methods covered in the lesson). Compose a plan consisting of the following parts. Present your plan on one whole sheet of short bond paper (maybe handwritten or typewritten).

   a. Name of the soil conservation method (2 pts)
   b. Materials to be used (3 pts)
   c. Procedure/steps presented in a flowchart (10 pts)
   d. Sketch showing how the materials used in the method are organized (10 pts)
   e. Description of the principle or mechanism by which the method will hold the soil (5 pts)

Assessment

Directions: Analyze the following images. Identify and describe the key feature(s) of the erosion control method shown in each image. Write your answer in the box provided. (10 pts)



Method:	Method:	Method:
Feature(s):	Feature(s):	Feature(s):

Method:	Method:
Feature(s):	Feature(s):

*Source:
https://education.hcswcd.org/uploads/1/5/4/8/15484824/dirt_secrets_in_the_soil_2.pdf
**for all five images*

Day 2 Waste Classification and Generation

Most of the wastes that are produced by humans are from **residential, industrial, and commercial activities**. These wastes are classified as **biodegradable**, **non-biodegradable**, **hazardous,** and **non-hazardous**, depending on their distinct physical and chemical properties.

At the end of this unit, you are expected to:

1. describe how people generate different types of waste (solid, liquid, and gaseous) as they use various materials and resources in everyday life. **(S11ES-Ii-19)**

What's In

Soil erosion brought about by agents like water and wind can be controlled by applying different methods including contour plowing/farming, crop rotation, mulching, no-till planting, strip-cropping, terracing, and windbreaks.

What is It

Waste is any unwanted or unusable material that is discarded after primary use. The United Nations Environment Program (UNEP) further defines waste as "substances or objects, which are disposed of are intended to be disposed of, or are required to be disposed of by the provisions of national law".

Humans generate tons and tons of waste daily as they engage in various activities. These wastes can be classified as **biodegradable**, **non-biodegradable**, **hazardous**, or **non-hazardous** based on their distinct physical and chemical properties.

Waste Classification:

- **Biodegradable Wastes** – Originate from plant and animal sources which may be decomposed or broken down by other living organisms (usually microorganisms)

- **Non-biodegradable Wastes** – Originate from inorganic or synthetic materials (non-living matter) which cannot be decomposed or broken down by living organisms
- **Hazardous wastes** – Exhibit properties of flammability/ignitability, reactivity, corrosivity, toxicity, and/or infectivity; pose a hazard to humans and the environment
- **Non-hazardous Wastes** – Do not exhibit the properties of hazardous waste but usually pose disposal problems in terms of their accumulated volume
- **Biomedical wastes** – Solid or liquid wastes, including containers, intermediate or end products generated by diagnosis, treatment, and research activities of medical sciences
- **Chemical wastes** – Represent hazardous fraction of industrial wastes (may also be generated in the household); require special handling to prevent dispersal of the wastes

Waste Generation:
*Transcript of the Youtube video "Source of Waste"
Source: https://www.youtube.com/watch?v=7S8VTbezuzw&t=18s

What's More

Directions: Fill out Tables 1-3 based on the concepts presented in the previous section. (12 pts)

Table 1. Biodegradable and Non-biodegradable Wastes

Classification	Characteristics	Examples
1. Biodegradable		
2. Non-biodegradable		

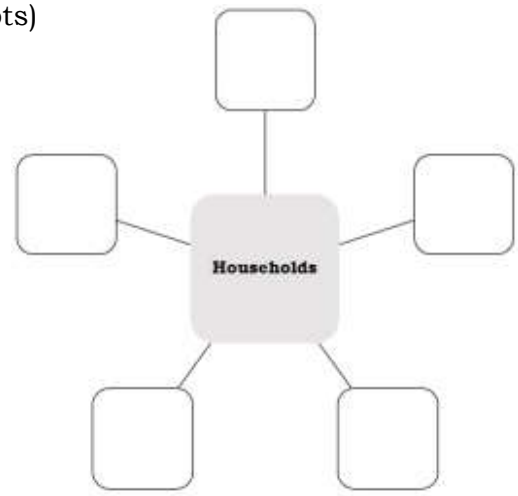
Table 2. Hazardous and Non-hazardous Wastes

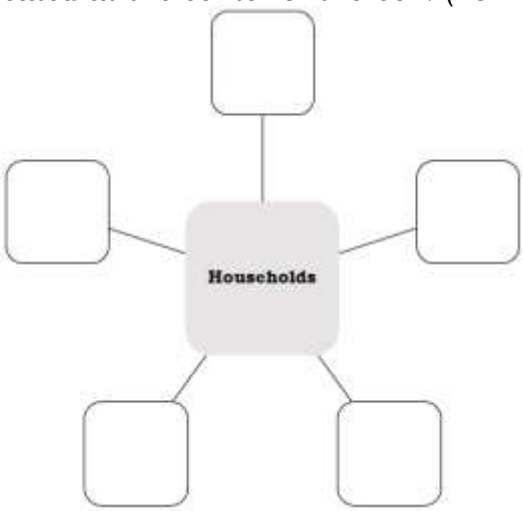
Classification	Characteristics	Examples
1. Hazardous		
2. Non-hazardous		

Table 3. Common Hazardous Wastes

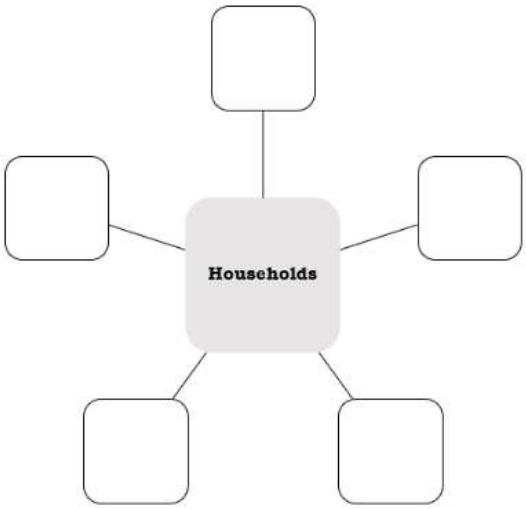
Classification	Characteristics	Examples
1. Chemical		
2. Biomedical		

Directions: Fill out the empty boxes in Schematic Charts 1-3 with names of sample wastes generated by the SOURCE indicated at the center of the box. (15 pts)





Schematic Chart 1.
Wastes from households (residential) (industrial)



Schematic Chart 2.
Wastes from factories

Guide Questions:

1. How would you know if a given waste is: (4 pts)
   1.1. biodegradable?
   1.2. non-biodegradable?
   1.3. hazardous?
   1.4. non-hazardous?
2. Why are chemical wastes and biomedical wastes considered HAZARDOUS? (1 pt)
3. Why is there a need to classify wastes? (1 pt)
4. Describe briefly how wastes are generated in households, factories, and retail establishments. (3 pts)

Schematic Chart 3.
Wastes from retailers (commercial)

What I Have Learned

Directions: Briefly answer the given questions below.

1. How are wastes classified? (2 pts)
2. How are different wastes generated? (3 pts)

What I Can Do

Directions: Classify the following wastes in the chart that follows. (20 pts)

Leaves, Milk jug, Mayonnaise jar, Jar metal lid, Used notepad sheet, Spray insecticide can, Vegetable scraps, Cereal carton box, Microwaveable plastic container, Soda can, Nail polish, Newspaper, Styrofoam cup, Uprooted weeds, a Broken pencil with eraser, Oil-based paint can, Grass clippings, Shampoo sachets, Tin can, Old leather shoe

Can be Recycled	Can Be Composted	Hazardous	Must be Landfilled

Note: A **landfill** is a place where wastes are disposed and buried by covering with soil; **sanitary landfills** contain wastes originating from daily human activities (municipal solid wastes) while **secure landfills** contain hazardous wastes isolated from people, wildlife, and ecosystems.

Assessment

Part A - Directions: In the chart that follows, put a check mark/s under the APPROPRIATE column for each waste item indicated. (5 pts)

Waste	Biodegradable	Non-biodegradable	Hazardous	Non-hazardous
1. Rubber tire				
2. Spoiled food				
3. Animal carcasses				
4. PET bottle				
5. Used syringe				

Part B - Directions: Describe how wastes are generated in each of the following human activities. (15 pts)

1. Going into a supermarket to buy groceries
2. Manufacture of rubber shoes
3. Cooking foods in fast food restaurant

REFERENCES

https://www.classzone.com/science_book/mls_grade7_FL/248_252.pdf .
https://www.youtube.com/watch?v=yBAgWDZH6_c&list=PL5V9GXQlS0_v-GNP3Ck8h-VBHeIyo3dPv&index=4 (accessed June 13, 2020).
Extension, Utah State University.
https://utah.agclassroom.org/teacher/matrix/lessonplan.cfm?lpid=22&search_term_lp=soil (accessed June 13, 2020).
https://www.classzone.com/science_book/mls_grade7_FL/248_252.pdf (accessed June 13, 2020).
https://www.wsfcs.k12.nc.us/cms/lib/NC01001395/Centricity/Domain/7110/Preventing%20Soil%20Erosion%20WS.pdf (accessed June 13, 2020).
https://slideplayer.com/slide/13010576/ (accessed June 10, 2020).
https://www.slideshare.net/anshikamehrotra7/generation-of-waste-71694726 (accessed June 11, 2020)
https://www.youtube.com/watch?v=7S8VTbezuzw&t=18s (accessed June 10, 2020)

Waste Generation and Management
Quarter 1 – Week 7

What I Need to Know

This module was designed and written with you in mind. It helps you master the topics about waste generation and management. The scope of this module permits its uses in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course, but the order in which you read them can be changed to correspond with the textbook you are now using.
This module will focus on waste generation & management.

After going through this module, you are expected to:

1. Compare the decomposition rate of solid wastes.
2. Explain how waste can affect our environment.

What I know

LET'S TRY! (PRE-ASSESSMENT)

Directions: Fill in the Frayer's model below to assess your prior knowledge and understanding of the waste generation and management.

Define what is waste management	**How waste is generated**
Type of waste management practices	**Ways to reduce waste in the community**

WASTE

Day 1 & 2 Waste Generation & Management

Waste can either be in solid or in liquid form. It generates from the products which are discarded after its primary use. The Philippines generates more solid waste as our population continues to rise especially if the standard of living of the people living in the urban and rural places is improving. Based on Senate Economic Planning Office, our country generates 40,087.45 tons of waste per day last 2016. At present we should learn the importance of waste management to help control and lessen our concern in the increasing production of it. Topics about waste management have been widely discussed across the school curriculum not just in the Philippines but also in other countries. However, why waste generation and management are still a major concern we are facing today?

At the end of this lesson, you are expected to:

1. explain how different types of waste affect people's health and the environment.
   [S11ES-Ii-j-20]

What's In

Most of the wastes that are produced daily result from residential, industrial, and commercial activities. The growing world population is causing negative impacts on the planet. The current model of production and consumption generates a lot of waste that, in many cases, does not get reused or recycled.

A few years ago, waste management focused mainly on sending it to tips to pile up or to **incinerators**. This solution is unsustainable for several reasons:

- It carries critical **risks for the environment**, living beings, and human health.
- It does not **reduce the consumption of resources** (raw materials and energy)
- It does not act on the cause (the consumption model), only on the consequence (**waste management**).

What is It

Waste Management

Linear v. circular production mode

A Linear economic model to a circular economy emulates nature by converting waste into resources. This type of economy promotes a decrease in consumption and considering a product's usefulness and capability to be reused or recycled. And to implement the circular economy model, it is necessary to change the mentality of companies and consumers. Companies, for their part, need to adapt **product design**

according to circular economy principles, using waste as raw materials and reducing non-reusable products.

The circular economy production model advocates using as many biodegradable materials as possible in manufacturing consumer goods – biological nutrients – so that these can return to nature without causing environmental damage once their useful lives are over.

Consumers and the 3 Rs rule

Reduce implies preventing the formation of waste: modifying production models, our consumption habits and acquiring products responsibly. **Reusing** signifies using a product again for the same function (repairing it) or an alternative use, lengthening its useful life. Finally, **Recycle** permits us to use the different materials in waste and introduce them into the production cycle as raw materials.

What's More

Materials: Worksheets, Pen, Large Plastics Bag & Hand Gloves

Procedure:

1. Collect waste products at home. (Note: disinfect the waste and wear hand gloves and face masks before doing the activity.)
2. Identify and list all the waste that can be burned.
3. Identify and list all the waste that can be recycled.
4. Identify and list all the waste that can decompose rapidly when burned.
5. Record the total number of pieces of waste you have collected.

Total number of pieces _____

Type of waste	Combustible	Recyclable	Decomposable or Biodegradable
Number of pieces			
Percentage			

To compute the percentage of waste per group, use the formula

$$\% = \frac{\text{Number of pieces per group}}{\text{Total number of pieces}} \times 100$$

Guided Questions:

1. Which group of waste was the most common? Why do you think so?
2. Based on your collection, what waste is the most difficult to discard?
3. Based on the activity what have you found out in general?

What I have Learned

Directions: Briefly answer the given question below.

What are the best waste management practices you have learned on this topic? Discuss your insights on how it can be implemented in your community.

What I can do

Directions: Read the article and answer the corresponding guide questions.

Waste segregation more crucial amid pandemic

By: Jhesset O. Enano - Philippine Daily Inquirer / 04:54 AM May 20, 2020

You may access the article through this link: https://newsinfo.inquirer.net/1277808/waste-segregation-more-crucial-amid-pandemic#ixzz6OraV2Ktv

Guided Questions:

1. Based on the article, how can we relate the COVID 19 pandemic in waste production?
2. Does incinerating infectious waste a good idea? Why and why not?
3. What are your household waste management practices during the enhanced community quarantine period?

Assessment

Directions: Write True if the statement is correct and False if not. Write your answer in the space provided.

_____1. Using reusable bags while shopping can help reduce waste.
_____2. The Linear production model promotes the use of 3Rs to the consumers.
_____3. Recycling implies preventing the formation of waste: modifying production models, our consumption habits, and acquiring products responsibly.
_____4. You are recycling if you make a new basket from an old ice cream bucket.
_____5. Incineration of garbage can cause a serious problem to our environment.

Additional Activities

Based on what you have learned from this week's lesson, you are task to create a vlog wherein you will present your proposed community waste management plan.

REFERENCES

Sara Mae D. Mawis, "Solid waste mismanagement in the Philippines." May 18, 2019. https://business.inquirer.net/270819/solid-waste-mismanagement-in-the-philippines

Acciona, "Waste Generation and Management" Accessed on June 5, 2020. https://www.activesustainability.com/environment/waste-generation-management/

Jhesset O. Enano, "Waste segregation more crucial amid pandemic." May 20, 2020. https://newsinfo.inquirer.net/1277808/waste-segregation-more-crucial-amid pandemic#ixzz6OraV2Ktv

Lilia G. Vengco, Teresita F. Religiosos, Delia Cordero-Navanza, *You and the Natural World of Science 7* (Phoenix Publishing House, Inc., 2013), 225- 227

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